Lab 10

MongoDB multi-document transactions | Docker demonstration submission

Lab 10.1	Lab 10.2
Bank transfer using sessions, insufficient-funds rollback, transfer history, and exponential retry for transient transaction errors.	Concurrent inventory reservations with conditional stock updates, atomic pending orders, contention handling, and throughput output.

Run the demonstration

Start the application and single-node MongoDB replica set from the project folder:

```
docker compose up --build -d
docker compose run --rm app python scripts/lab10_transactions.py all
```

Compare inventory runs using --workers 1, 4, 8, or 16. Full instructions are in docs/lab10.md.

Published Docker image

Docker Hub image: krish1226/lab10:lab10
The image push completed successfully. Pull it using the command below.

Repository: https://hub.docker.com/r/krish1226/lab10

```
docker pull krish1226/lab10:lab10
```

Project files: Dockerfile, docker-compose.yml, scripts/lab10_transactions.py, and docs/lab10.md. Database credentials are supplied at runtime and are not built into the image.